Supplementary file 5. Effects and outcomes of interventions on response to VAW during routine SRH consultation plus community engagement

Outcomes	Results	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Direct effects on health-related cognition and emotions						
HCP knowledge about VAW and relevant procedures (n=2)						
Proportion who knew whether there was law that deals with family violence	Pre-intervention = 71% Post-intervention = 90%	19%	?HCPs (UBA/3 years)	Bott 2014[30]	High	Systems level activities, HCP training and ongoing support, infrastructure improvement, integrated family planning-GBV consultation, referral to onsite GBV specialist, community education contributed towards 19% increase in HCP knowledge about legal side of VAW
Proportion who could explain legal obligation of providers regarding family violence	Pre-intervention = 14% Post-intervention = 69%	55%				
Proportion (n) who knew the main objectives of the programme	100% (35/35)		35 HCPs (Cross sectional/4-year service data)	Sithole 2018[53]	High	HCP training, infrastructure improvement, community education on post-rape care contributed towards 25% to 100% HCP awareness about post-rape care
Proportion of doctors who knew the tools to monitor the programme	25% (1/4)					
Proportion of doctors who knew the correct treatment guidelines	25% (1/4)					
Proportion of doctors who did not know the management process	75% (3/4)					
Proportion of nurses who knew the management process	100% (27/27)					
HCP attitudes about VAW (n=1)						
Reduction in proportion who blamed victims of physical and sexual violence (5 indicators), pre-post-intervention	-	By 29% for women provoke physical aggression By 13% for men cannot control their sexual behaviour	?HCPs (UBA/3 years)	Bott 2014[30]	High	Systems level activities, HCP training and ongoing support, infrastructure improvement, integrated family planning-GBV consultation, referral to onsite GBV specialist, community education contributed towards 13-29% reduction in negative attitudes about GBV among HCPs
HCP readiness for identifying and responding to VAW (n=1)						
Reduction in proportion of cited 9 barriers to identifying IPV, pre-post intervention	-	By 29% for cultural divide between client and provider By 3% for time constraints	?HCPs (UBA/3 years)	Bott 2014[30]	High	Systems level activities, HCP training and ongoing support, infrastructure improvement, integrated family planning-GBV consultation, referral to onsite GBV specialist, community education contributed 3% to 29% reduction in perceived barriers to identifying GBV among HCPs

Outcomes	Results	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Increase in proportion who felt prepared to provide counselling about emergency contraception to GBV victims	-	By 96% for counselling about emergency contraception				Systems level activities, HCP training and ongoing support, infrastructure improvement, integrated family planning-GBV consultation, referral to onsite GBV specialist, community education contributed towards 95% increase in HCP preparedness to identify and respond to GBV patients
Women attitudes about VAW (n=1)						
Proportion (n) who justified husband physical abuse because of childcare, post-intervention	Intervention = 41.8% (261/625) Control = 45.5% (284/624)	OR = 0.81 (0.60; 1.09)	656 women (cluster RCT/28 months)	Settergren 2018[60]	High	Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education improved 1 out of 5 indicators of women's accepting attitudes towards VAW
Proportion (n) who justified husband physical abuse because she refuses to have sex with her partner, post-intervention	Intervention = 21.0% (131/625) Control = 23.7% (148/624)	OR = 0.65 (0.46; 0.91)				
Mean (SD) score of the Violence domain of the Gender Equitable Men Scale, post-intervention	Intervention = 13.17 (3.98) Control = 12.51 (3.93)	MD = 1.08 (0.52; 1.65)				Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education improved women's attitudes towards more equitable gender roles
Mean (SD) score of the Domestic chores and daily life domain domain of the Gender Equitable Men Scale, post-intervention	Intervention = 8.74 (3.63) Control = 7.62 (3.14)	MD = 1.26 (0.81; 1.71)				
Intermediate effects on health-related behaviour and ptractices						
VAW enquiry rate (n=2)						
Proportion (n) who received GBV screening and counselling, post-intervention	Intervention = 88.5% (1251/1413) Control = 91/7% (442/482) p=0.785	-	656 women (cluster RCT/28 months)	Settergren 2018[60]	High	Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education had no effect on GBV enquiry rate
Proportion (n) who were screened for IPV of those attended clinic	94.5% (171/181)	-	171 women (Cross-sectional/4 months)	Samandari 2016[49]	High	System level activities, 7-day HCP training and ongoing support, integrated family planning-IPV consultation, external referral, community education contributed towards 95% IPV enquiry rate
Provision of VAW referrals (n=2)						
Proportion (n) who were referred to safe house and shelter of those screened, post-intervention	Intervention = 12.3% (173/1412) Control = 2.3% (11/488) p=0.216	-	656 women (cluster RCT/28 months) (cluster RCT/28 months)	Settergren 2018[60]	High	Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education

Outcomes	Results	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
						had no effect on rates of referrals to safe house and shelter
Proportion (n) who were signposted to IPV services of those disclosed	100% (157/157)	-	171 women (Cross-sectional/4 months)	Samandari 2016[49]	High	System level activities, 7-day HCP training and ongoing support, integrated family planning-IPV consultation, external referral, community education contributed towards 100% signposting to IPV services
Provision of safety planning (n=1)						
Proportion (n) who received safety planning of those disclosed IPV	87.3% (137/157)	-	171 women (Cross-sectional/4 months)	Samandari 2016[49]	High	System level activities, 7-day HCP training and ongoing support, integrated family planning-IPV consultation, external referral, community education contributed towards 87% safety planning rate
Provision of post-rape care (n=3)						
Mean number of rape cases presenting to hospital per month.	Pre-intervention = 8 Post-intervention = 13	-	334 survivors of sexual assault (UBA and cross-sectional/not reported)	Kim 2007[42]	High	Systems level activities, 2-day HCP training, infrastructure improvement, community education on post-rape care contributed towards increased number of rape cases presenting to hospital
Proportion (n) of eligible patients who received post-rape medical kit	100% (2,081/2,081)	-	13 sites, 2081 patients (Cross-sectional/4-year service data)	Bress 2018[31]	High	HCP training and ongoing support, infrastructure improvement, community education on post-rape care contributed towards 100% provision of post-rape care medical kit
Change in proportion who attended within 72 hours, over 4 years	-	46%	80 HCPs, 1669 patients (Cross-sectional/4 years)	Sithole 2018[53]	High	HCP training, infrastructure improvement, community education on post-rape care contributed towards improvement on 6 indicators of post-rape care provision
Change in proportion who received HIV post-exposure prophylaxis, over 4 years	-	31%				
Change in proportion who received counselling, over 4 years	-	65%				
Change in proportion who received HIV testing, over 4 years	-	96.4%				
Change in proportion who received emergency contraception, over 4 years	-	8%				
Change in proportion who received STI prophylaxis	-	26%				
VAW disclosure rates (n=2)						
Proportion who disclosed GBV of those screened	14%	-	? women (UBA/3 years)	Bott 2004[30]	High	Systems level activities, HCP training and ongoing support, infrastructure improvement, integrated family planning-GBV consultation, referral to onsite GBV specialist, community education contributed towards 14% GBV identification rate

Outcomes	Results	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Proportion (n) who disclosed IPV of those screened	91.8% (157/171)	-	171 women (Cross-sectional/4 months)	Samandari 2016[49]	High	System level activities, 7-day HCP training and ongoing support, integrated family planning-IPV consultation, external referral, community education contributed towards 92% IPV identification rate
Proportion (n) who disclosed GBV of those screened	37% (49/134)	-	134 women (Cross-sectional/5 months)	Turan 2013[56]	High	40-hour HCP training, integrated ANC-GBV consultation, assisted external referral, community education contributed towards 37% IPV identification rate
VAW referrals uptake (n=2)						
Proportion who took referral of those disclosed GBV	30%	-	? women (UBA/3 years)	Bott 2004[30]	High	Systems level activities, HCP training and ongoing support, infrastructure improvement, integrated family planning-GBV consultation, referral to onsite GBV specialist, community education contributed towards 30% uptake of referrals
Proportion (n) who took referral of those disclosed IPV	0.6% (1/157)	-	171 women (Cross-sectional/4 months)	Samandari 2016[49]	High	System level activities, 7-day HCP training and ongoing support, integrated family planning-IPV consultation, external referral, community education contributed towards 0.6% uptake of external referrals
Proportion (n) who took referral of those disclosed GBV	53% (26/49)	-	134 women (Cross-sectional/5 months)	Turan 2013[56]	High	40-hour HCP training, integrated ANC-GBV consultation, assisted external referral, community education contributed towards 53% uptake of referrals
Use of SRH services (n=1)						
Proportion (n) who attended any antenatal care, post-intervention	Intervention = 88.7% (1597/1800) Control = 82.4% (1526/1851)	RD = 0.063 (-0.044; 0.170)	1837 women (cluster RCT/12 months)	Cockcroft 2019[34]	Low	HCP training, infrastructure improvement, integrated DV-universal home visits that discussed domestic violence, heavy work in pregnancy, ignorance of danger signs, and lack of spousal communication with pregnant women and their spouses had no effect on 8 indicators of use of antenatal care, institutional delivery, or skilled birth attendance
Proportion (n) who delivered in a health facility, post-intervention	Intervention = 30.1% (475/1579) Control = 21.9 (391/1785)	RD = 0.082 (-0.071; 0.235)				
Proportion (n) who delivered by a skilled health worker post-intervention	Intervention = 29.3 (463/1579) Control = 22.7 (404/1783)	RD = 0.067 (-0.081; 0.214)				
Mean (SD of GBV client visits per facility, post-intervention	Intervention = 237.8 (110.58) Control = 81.5 (46.09) p=0.010	-	656 women (cluster RCT/28 months)	Settergren 2018[60]	High	Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education

Outcomes	Results	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
						increased use of SRH services by women with experience of GBV
Proportion (n) of GBV clients who received HIV testing and counselling, post-intervention	Intervention = 73.3% (1038/1416) Control = 20.9% (102/488) p=<0.001	-				Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education increased use of 4 (HIV testing and counselling, family planning, forensic exam) out of 24 SRH services
Proportion (n) of GBV clients who received STI test, post-intervention	Intervention = 21.8% (308/1415) Control = 11.5% (56/488) p=0.128	-				
Safety behaviour (n=1)						
Proportion (n) of women who used condom, post-intervention	Intervention = 17.0% (157/931) Control = 16.0% (192/1170)	PRR = 1.03 (0.85; 1.25) aPRR = 1.01 (0.84; 1.21)	6 facilities, 1812 women (cluster RCT/35 months)	Wagman 2015[59]	High	HCP training, integrated HIV-IPV consultation, onsite referral, community education had no effect on 6 indicators of risk behaviours among women
Health outcomes						
Re-exposure to VAW (n=3)						
Proportion (n) who did not experience physical domestic violence during pregnancy, post-intervention	Intervention = 97.4 (1772/1820) Control = 90.9 (1677/1844)	RD = 0.064 (0.045; 0.084)	1837 women (RCT/12 months)	Cockcroft 2019[34]	Low	HCP training, infrastructure improvement, integrated DV-universal home visits that discussed domestic violence, heavy work in pregnancy, ignorance of danger signs, and lack of spousal communication with pregnant women and their spouses reduced physical domestic violence
Proportion (n) who experienced any IPV in past 12 months, post-intervention	Intervention = 37.2% (207/556) Control = 45.7% (268/587)	OR = 0.85 (0.62; 1.16)	6 facilities, 735 households (cluster RCT/28 months)	Settergren 2018[60]	High	Systems level activities, HCP training, infrastructure improvement, integrated HIV-GBV consultation, onsite and external referral, community and couple education had no effect on any IPV
Proportion (n) who experienced physical IPV in past 12 months, post-intervention	Intervention = 12% (217/1812) Control = 16% (346/2127)	PRR = 0.74 (0.63; 0.86) aPRR = 0.79 (0.67; 0.92)	6 facilities, 1812 women (cluster RCT/35 months)	Wagman 2015[59]	High	HCP training, integrated HIV-IPV consultation, onsite referral, community education reduced incidents of physical and sexual IPV and had no effect on emotional IPV.
Proportion (n) who experienced emotional IPV in past 12 months, post-intervention	Intervention = 18% (311/1737) Control = 20% (409/2039)	PRR = 0.89 (0.78; 1.02) aPRR = 0.91 (0.79; 1.04)				
Proportion (n) who experienced sexual IPV in past 12 months, post-intervention	Intervention = 10% (167/1737) Control = 13% (261/2038)	PRR = 0.75 (0.62; 0.90) aPRR = 0.80 (0.67; 0.97)				
Sexual and reproductive health (n=1)						

Outcomes	Results	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Proportion (n) who did not have swelling of face or hands, post-intervention	Intervention = 97.4% (1790/1837) Control = 71.1% (1317/1853)	RD = 0.264 (0.194; 0.333)	1837women (RCT/12 months)	Cockcroft 2019[34]	Low	HCP training, infrastructure improvement, integrated DV-universal home visits that discussed domestic violence, heavy work in pregnancy, ignorance of danger signs, and lack of spousal communication with pregnant women and their spouses improved 9 out of 13 indicators of pregnancy and postpartum complications
Proportion (n) who did not have raised blood pressure, post-intervention	Intervention = 96.6% (1409/1458) Control = 85.1% (1269/1492)	RD = 0.116 (0.042; 0.190)				
Proportion (n) who did not have post-partum sepsis, post-intervention	Intervention = 81.1% (1478/1822) Control = 48.8% (903/1852)	RD = 0.324 (95% CI 0.115; 0.493)				
Incidence of HIV per100 person-years	Intervention = 0.99 Control = 1.15	IRR = 0.86 (0.61; 1.22 aIRR = 0.72 (0.49; 1.07))	6 facilities, 1812 women (cluster RCT/35 months)	Wagman 2015[59]	High	HCP training, integrated HIV-IPV consultation, onsite referral, community education had no effect on incidence of HIV among women

Note. VAW violence against women. GBV gender-based violence. DV domestic violence. IPV intimate partner violence. SRH sexual and reproductive health. STI Sexually transmitted infections. HIV human immunodeficiency virus. ANC antenatal care. HCP health care provider. RCT randomised controlled trial. CBA controlled before-after study. UBA uncontrolled before-after study. CI confidence interval. SD standard deviation. IQR interquartile range. MD mean difference. OR odds ratio. RD risk difference. IRR incidence rate ratio.